



# Technoverse Hackathon

## 2026



# Business Plan

## WHY

### *Explain the Problem*

**Problem Description & Business Scenario :** Healthcare systems generate vast patient data, but it is designed for clinicians. As a result, many patients struggle to understand medical reports due to complex terminology and lack of personalized explanations.

A significant number of patients miss recovery opportunities due to lack of awareness, while others develop complications because of improper medication usage and misunderstanding of their condition.

Although hospitals attempt to educate patients through pamphlets, advertisements, and in-hospital materials, these methods are often ignored or ineffective.

Additionally, medical information remains fragmented across hospitals, labs, and physical documents, There is no secure and centralized platform for patients to manage their medical records.

With the rise of digital health and chronic diseases, there is a growing demand for patient-centric, engaging, and easy-to-understand healthcare solutions, yet a major gap still exists in health literacy and patient education.

**Problem Scope:** Focuses on the gap in patient understanding of medical conditions and reports, not on clinical diagnosis or treatment decisions

Limitations: Dependent on the quality and completeness of uploaded medical data

**Target Users/Stakeholders: Patients**



# Business Plan

**Solution Overview:** CareCompanion is an AI-powered patient education and health management platform designed to simplify medical information and improve patient understanding. It transforms complex medical reports into easy-to-understand explanations and personalized educational videos, enabling patients to better understand their condition and follow treatment plans effectively.

CareCompanion also includes a secure centralized document vault where patients manage all their medical records.

An integrated AI assistant, the “Director”, answers patient queries, and guides users through their reports and treatment understanding.

## Technical Details:

**Frontend:** React.js

**Backend:** Fast API (Python)

**Database & Auth:** Supabase (secure storage, authentication, RBAC)

**AI & Processing Layer:** LLM (Llama 3.1 via Groq API)

**Document Processing:** PyPDF2 / OCR tools

**Media Generation Pipeline:** FLUX.2, Wan 2.2, Ffmpeg.

**Innovation:** Transforms traditional healthcare interaction by converting medical reports into personalized AI-generated educational videos,  
Introduces an education-first AI approach, focusing on patient understanding rather than only diagnosis

**Market Potential:** The global digital health market is rapidly expanding, valued at ~\$347 billion in 2025 and expected to reach ~\$946 billion by 2030 with strong growth (~22% CAGR)

# How



# Bussiness Plan

## Why are the technologies you used are appealing for the solution:

### **React + FastAPI:**

Enables fast, scalable, and full-stack development. FastAPI is a modern, high-performance Python framework Using FastAPI allows seamless future integration of AI/ML models for disease classification and analysis, aligning with long-term scalability..

### **LLMs (Llama 3.1 via Groq API):**

Groq provides high request throughput, which is crucial for building responsive AI-driven applications at scale.

### **AI Media Tools (Flux2-klein, Wan 2.2, FFmpeg):**

Open-source tools that enable conversion of medical data into engaging video-based content, while significantly reducing development and operational costs.

### **Supabase:**

Provides a secure, scalable backend with authentication and database management. Also offers exposure to industry-standard SQL-based systems, ensuring practical and production-ready architecture.n.

## WHAT

### Key Benefits (Quantified Impact)

- **Better Understanding** → Simplified reports & visuals
- **Time Saved** → Faster consultations
- **20–25% Cost Reduction** → Fewer follow-ups
- **Scalable** → Supports millions of users
- **Better Health Outcomes** → Reduced complications



# Financials & Timelines

## Investments

Cloud & API: ₹5K/month

2–3 Developers

Open-source tools

## Returns

Improved patient understanding

Revenue via subscriptions

Scalable healthcare platform

Risk if not solved: poor awareness

## Timelines

Phase 1: UI + Upload (1 month)

Phase 2: AI processing (2 months)

Phase 3: Video + Assistant

Phase 4: Deployment

# Thank you